Supplementary Material for

A Bayesian Critique of Rank-Based Methods for Surrogate Marker Evaluation by Carlotti and Parast

A Derivation of θ and δ as Expectations

Under the model in Equation (6), it is easy to see that

$$Y_{1i} - Y_{0i} \mid X_i = x \overset{\text{ind.}}{\sim} \mathcal{N}\left(\frac{\mu^x_{Y_1} - \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right), \quad i = 1, \ldots, n, \ \forall\, x \in \mathcal{X}. \tag{A.1}$$

Therefore, we have that

$$\begin{aligned}
V^x_Y &= \mathbb{P}\left(Y_{1i} > Y_{0i} \mid X_i = x\right) \\
&= \mathbb{P}\left(Y_{1i} - Y_{0i} > 0 \mid X_i = x\right) \\
&= \mathbb{P}\left(\frac{Y_{1i} - Y_{0i} - \mu^x_{Y_1} + \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}} > \frac{-\mu^x_{Y_1} + \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}} \mid X_i = x\right) \\
&= \mathbb{P}\left(Z > \frac{-\mu^x_{Y_1} + \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right) \\
&= 1 - \Phi\left(\frac{-\mu^x_{Y_1} + \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right) \\
&= \Phi\left(\frac{\mu^x_{Y_1} - \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right), \ \forall\, x \in \mathcal{X},
\end{aligned} \tag{A.2}$$

where $Z \sim \mathcal{N}(0,1)$ and Φ is the standard normal cumulative distribution function. Then, by the law of total probability, we can expand V_Y as follows:

$$\begin{aligned}
V_Y &= \mathbb{P}\left(Y_{1i} > Y_{0i}\right) \\
&= \int_{\mathcal{X}} V^x_Y\, f(x)\, \mathrm{d}x \\
&= \int_{\mathcal{X}} \Phi\left(\frac{\mu^x_{Y_1} - \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}}\right) f(x)\, \mathrm{d}x.
\end{aligned} \tag{A.3}$$

Since the same reasoning applies to V_S, we obtain:

$$\theta = \int_{\mathcal{X}} \left(\Phi \left(\frac{\mu^x_{Y_1} - \mu^x_{Y_0}}{\sqrt{\Sigma^x_{Y_1Y_1} + \Sigma^x_{Y_0Y_0} - 2\Sigma^x_{Y_1Y_0}}} \right) - \Phi \left(\frac{\mu^x_{S_1} - \mu^x_{S_0}}{\sqrt{\Sigma^x_{S_1S_1} + \Sigma^x_{S_0S_0} - 2\Sigma^x_{S_1S_0}}} \right) \right) f(x)\,\mathrm{d}x. \tag{A.4}$$

To derive an analytical expression for δ, we start by noting that since

$$Y_{1i} - Y_{0j} \mid X_i = x_i, X_j = x_j \sim \mathcal{N} \left(\frac{\mu^{x_i}_{Y_1} - \mu^{x_j}_{Y_0}}{\sqrt{\Sigma^{x_i}_{Y_1Y_1} + \Sigma^{x_j}_{Y_0Y_0}}} \right), \quad \forall\, x_i, x_j \in \mathcal{X},\ i,j = 1,\ldots,n, \tag{A.5}$$

then we have that

$$\begin{aligned} U_Y^{x_ix_j} &= \mathbb{P}\left(Y_{1i} > Y_{0j} \mid X_i = x_i, X_j = x_j\right) \\ &= \mathbb{P}\left(\frac{Y_{1i} - Y_{0j} - \mu^{x_i}_{Y_1} + \mu^{x_j}_{Y_0}}{\sqrt{\Sigma^{x_i}_{Y_1Y_1} + \Sigma^{x_j}_{Y_0Y_0}}} > \frac{-\mu^{x_i}_{Y_1} + \mu^{x_j}_{Y_0}}{\sqrt{\Sigma^{x_i}_{Y_1Y_1} + \Sigma^{x_j}_{Y_0Y_0}}} \mid X_i = x_i, X_j = x_j \right) \\ &= \mathbb{P}\left(Z > \frac{-\mu^{x_i}_{Y_1} + \mu^{x_j}_{Y_0}}{\sqrt{\Sigma^{x_i}_{Y_1Y_1} + \Sigma^{x_j}_{Y_0Y_0}}} \right) \\ &= 1 - \Phi\left(\frac{-\mu^{x_i}_{Y_1} + \mu^{x_j}_{Y_0}}{\sqrt{\Sigma^{x_i}_{Y_1Y_1} + \Sigma^{x_j}_{Y_0Y_0}}} \right) \\ &= \Phi\left(\frac{\mu^{x_i}_{Y_1} - \mu^{x_j}_{Y_0}}{\sqrt{\Sigma^{x_i}_{Y_1Y_1} + \Sigma^{x_j}_{Y_0Y_0}}} \right), \quad \forall\, x_i, x_j \in \mathcal{X},\ i,j = 1,\ldots,n. \end{aligned} \tag{A.6}$$

Then, by the law of total probability, we can expand U_Y as follows:

$$\begin{aligned} U_Y &= \mathbb{P}\left(Y_{1i} > Y_{0j}\right) \\ &= \int_{\mathcal{X}} \int_{\mathcal{X}} U_Y^{x_i,x_j} \, f(x_i) \, f(x_j) \, \mathrm{d}x_i \, \mathrm{d}x_j \\ &= \int_{\mathcal{X}} \int_{\mathcal{X}} \Phi\left(\frac{\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}} \right) f(x_i) \, f(x_j) \, \mathrm{d}x_i \, \mathrm{d}x_j. \end{aligned} \tag{A.7}$$

Since the same reasoning applies to U_S, we obtain:

$$\delta = \int_{\mathcal{X}} \int_{\mathcal{X}} \left(\Phi\left(\frac{\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}} \right) - \Phi\left(\frac{\mu_{S_1}^{x_i} - \mu_{S_0}^{x_j}}{\sqrt{\Sigma_{S_1S_1}^{x_i} + \Sigma_{S_0S_0}^{x_j}}} \right) \right) f(x_i) \, f(x_j) \, \mathrm{d}x_i \, \mathrm{d}x_j. \tag{A.8}$$

Denoting $\theta^x = V_Y^x - V_S^x$ and $\delta^{x_ix_j} = U_Y^{x_ix_j} - U_S^{x_ix_j}$, the above can be written compactly as:

$$\theta = \int_{\mathcal{X}} \theta^x \, f(x) \, \mathrm{d}x, \quad \delta = \int_{\mathcal{X}} \int_{\mathcal{X}} \delta^{x_ix_j} \, f(x_i) \, f(x_j) \, \mathrm{d}x_i \, \mathrm{d}x_j. \tag{A.9}$$

Under the additional assumption that $\Sigma_{Y_1Y_0}^x = \Sigma_{S_1S_0}^x = 0$ for all $x \in \mathcal{X}$, it follows that $V_Y^x = U_Y^{xx}$ and $V_S^x = U_S^{xx}$, so that $\theta^x = \delta^{xx}$. Defining

$$g(x_i, x_j) = \Phi\left(\frac{\mu_{Y_1}^{x_i} - \mu_{Y_0}^{x_j}}{\sqrt{\Sigma_{Y_1Y_1}^{x_i} + \Sigma_{Y_0Y_0}^{x_j}}} \right) - \Phi\left(\frac{\mu_{S_1}^{x_i} - \mu_{S_0}^{x_j}}{\sqrt{\Sigma_{S_1S_1}^{x_i} + \Sigma_{S_0S_0}^{x_j}}} \right), \tag{A.10}$$

we have $\delta^{x_ix_j} = g(x_i, x_j)$ and $\theta^x = g(x, x)$, so that

$$\delta = \mathbb{E}_{(X_i,X_j)\sim f^2}(g(X_i, X_j)), \quad \theta = \mathbb{E}_{X_i\sim f}(g(X_i, X_i)). \tag{A.11}$$

If the variances are further normalized so that $\Sigma^x_{Y_1Y_1} = \Sigma^x_{S_1S_1} = \Sigma^x_{Y_0Y_0} = \Sigma^x_{S_0S_0} = \frac{1}{2}$ for all $x \in \mathcal{X}$, then g simplifies to

$$g(x_i, x_j) = \Phi\left(\mu^{x_i}_{Y_1} - \mu^{x_j}_{Y_0}\right) - \Phi\left(\mu^{x_i}_{S_1} - \mu^{x_j}_{S_0}\right). \tag{A.12}$$

B Derivation of $|\theta - \delta|$ in the Binary Covariate Setting

Under the setting described in Section 2.4.1 of the main paper, let f = Bernoulli(1/2). Denoting $\theta^x = g(x, x)$ and $\delta^{x_i x_j} = g(x_i, x_j)$ for $x, x_i, x_j \in \{0, 1\}$, the expectation form of θ and δ reduces to

$$\theta = \frac{\theta^0 + \theta^1}{2}, \quad \delta = \frac{\delta^{00} + \delta^{11} + \delta^{01} + \delta^{10}}{4}, \tag{A.13}$$

which implies that

$$|\theta - \delta| = \frac{|\delta^{00} + \delta^{11} - \delta^{01} - \delta^{10}|}{4}. \tag{A.14}$$

This expression highlights that the discrepancy between the two parameters is driven by differences between comparisons within the same covariate group and comparisons across different covariate groups. Importantly, this effect does not arise from treatment effect heterogeneity but from differences in mean potential outcomes across covariate groups.

To derive the closed-form formula for $|\theta - \delta|$ given in Section 2.4.1, suppose moreover that S is a perfect surrogate for Y with homogeneous treatment effect:

$$\mu^0_{Y_1} - \mu^0_{Y_0} = \mu^0_{S_1} - \mu^0_{S_0} = \mu^1_{Y_1} - \mu^1_{Y_0} = \mu^1_{S_1} - \mu^1_{S_0} = \Delta > 0. \tag{A.15}$$

Using the normalized variance assumption $\Sigma^x_{Y_1Y_1} = \Sigma^x_{Y_0Y_0} = \Sigma^x_{S_1S_1} = \Sigma^x_{S_0S_0} = \frac{1}{2}$ and $\Sigma^x_{Y_1Y_0} =$

$\Sigma^x_{S_1 S_0} = 0$ for all $x \in \{0, 1\}$, we can compute each component of δ directly. Within-group comparisons give:

$$\begin{aligned} \delta^{00} &= \Phi(\Delta) - \Phi(\Delta) = 0, \\ \delta^{11} &= \Phi(\Delta) - \Phi(\Delta) = 0. \end{aligned} \tag{A.16}$$

For cross-group comparisons, the treatment means cancel and only the control-group means remain:

$$\begin{aligned} \delta^{01} &= \Phi\big(\Delta + \mu^0_{Y_0} - \mu^1_{Y_0}\big) - \Phi\big(\Delta + \mu^0_{S_0} - \mu^1_{S_0}\big)\,, \\ \delta^{10} &= \Phi\big(\Delta + \mu^1_{Y_0} - \mu^0_{Y_0}\big) - \Phi\big(\Delta + \mu^1_{S_0} - \mu^0_{S_0}\big)\,. \end{aligned} \tag{A.17}$$

Substituting $d_Y = \mu^0_{Y_0} - \mu^1_{Y_0}$ and $d_S = \mu^0_{S_0} - \mu^1_{S_0}$ and recalling that $\delta^{00} = \delta^{11} = 0$, we obtain:

$$|\theta - \delta| = \frac{1}{4}\left|\Phi(\Delta + d_Y) - \Phi(\Delta + d_S) + \Phi(\Delta - d_Y) - \Phi(\Delta - d_S)\right|. \tag{A.18}$$

C Derivation of θ and δ in the Continuous Covariate Setting

Under the setting described in Section **??** of the main paper, let $f = \mathcal{N}(m, s^2)$ and $\mu^x = x\,(\beta_{Y_1}, \beta_{S_1}, \beta_{Y_0}, \beta_{S_0})^\top$. We use the following standard identity throughout:

$$\mathbb{E}_{Z\sim\mathcal{N}(0,1)}(\Phi(a + bZ)) = \Phi\left(\frac{a}{\sqrt{1 + b^2}}\right), \quad \forall\, a, b \in \mathbb{R}. \tag{A.19}$$

Derivation of δ. Writing $X_i\beta_{Y_1} - X_j\beta_{Y_0} = m(\beta_{Y_1} - \beta_{Y_0}) + s(\beta_{Y_1}Z_i - \beta_{Y_0}Z_j)$ where $Z_i, Z_j \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$, and applying Equation (A.19) twice (once over Z_j, once over Z_i):

$$\begin{aligned} U_Y &= \mathbb{E}_{X_i,X_j}(\Phi(X_i\beta_{Y_1} - X_j\beta_{Y_0})) \\ &= \Phi\left(\frac{m(\beta_{Y_1} - \beta_{Y_0})}{\sqrt{1 + s^2(\beta_{Y_1}^2 + \beta_{Y_0}^2)}}\right). \end{aligned} \tag{A.20}$$

An analogous calculation for U_S yields:

$$\delta = \Phi\left(\frac{m(\beta_{Y_1} - \beta_{Y_0})}{\sqrt{1 + s^2(\beta_{Y_1}^2 + \beta_{Y_0}^2)}}\right) - \Phi\left(\frac{m(\beta_{S_1} - \beta_{S_0})}{\sqrt{1 + s^2(\beta_{S_1}^2 + \beta_{S_0}^2)}}\right). \tag{A.21}$$

Derivation of θ. Writing $X_i(\beta_{Y_1} - \beta_{Y_0}) = m(\beta_{Y_1} - \beta_{Y_0}) + s(\beta_{Y_1} - \beta_{Y_0})Z_i$ and applying Equation (A.19) over Z_i:

$$\begin{aligned} V_Y &= \mathbb{E}_{X_i}(\Phi(X_i(\beta_{Y_1} - \beta_{Y_0}))) \\ &= \Phi\left(\frac{m(\beta_{Y_1} - \beta_{Y_0})}{\sqrt{1 + s^2(\beta_{Y_1} - \beta_{Y_0})^2}}\right). \end{aligned} \tag{A.22}$$

An analogous calculation for V_S yields:

$$\theta = \Phi\left(\frac{m(\beta_{Y_1} - \beta_{Y_0})}{\sqrt{1 + s^2(\beta_{Y_1} - \beta_{Y_0})^2}}\right) - \Phi\left(\frac{m(\beta_{S_1} - \beta_{S_0})}{\sqrt{1 + s^2(\beta_{S_1} - \beta_{S_0})^2}}\right). \tag{A.23}$$

Supremum of $|\delta|$ when S is a perfect surrogate. Setting $\beta_{Y_1} - \beta_{Y_0} = \beta_{S_1} - \beta_{S_0} = \Delta > 0$ so that $\theta = 0$, we have:

$$|\delta| = \left|\Phi\left(\frac{m\Delta}{\sqrt{1 + s^2((\beta_{Y_0} + \Delta)^2 + \beta_{Y_0}^2)}}\right) - \Phi\left(\frac{m\Delta}{\sqrt{1 + s^2((\beta_{S_0} + \Delta)^2 + \beta_{S_0}^2)}}\right)\right|. \tag{A.24}$$

This is maximized when one of β_{Y_0}, β_{S_0} equals zero and the other diverges to $\pm\infty$. Setting $\beta_{Y_0} = 0$ and $|\beta_{S_0}| \to +\infty$ (or vice versa):

$$\sup_{\beta_{Y_0},\beta_{S_0}} \{|\delta|\} = \left|\Phi\left(\frac{m\Delta}{\sqrt{1+s^2\Delta^2}}\right) - 0.5\right|. \tag{A.25}$$

Maximizing further over Δ by taking $|\Delta| \to +\infty$:

$$\sup_{\beta_{Y_0},\beta_{S_0},\Delta} \{|\delta|\} = \left|\Phi\left(\frac{m}{s}\right) - 0.5\right|, \tag{A.26}$$

which can be arbitrarily close to 0.5 whenever $|m/s|$ is large.

D Regularity and Posterior Consistency

Definition 1 (Regularity of a Bayesian model)**.** *Let the Bayesian model be defined as*

$$\begin{aligned} D_i \mid \phi &\overset{i.i.d.}{\sim} f_\phi, \quad i = 1, \ldots, n, \\ \phi &\sim \pi, \end{aligned} \tag{A.27}$$

where D_i is the observed data for unit i, f_ϕ is the likelihood function parametrized by ϕ, and π is a prior distribution over the parameter ϕ. We say that the Bayesian model is regular if the following conditions hold:

- ***Kullback-Leibler support of the prior****: The prior π assigns positive mass to every Kullback-Leibler neighborhood of the true parameter ϕ^*, i.e., for every $\varepsilon > 0$,*

$$\pi\left(\{\phi : D_{\mathrm{KL}}(f_{\phi^*} \| f_\phi) < \varepsilon\}\right) > 0, \tag{A.28}$$

  where $D_{\mathrm{KL}}(f_{\phi^*} \| f_\phi)$ is the Kullback-Leibler divergence between f_{ϕ^*} and f_ϕ.

- ***Identifiability***: *The model is identifiable, i.e., for every $\phi \neq \phi'$, we have that $f_\phi \neq f_{\phi'}$.*

Theorem 1 (Schwartz's theorem (Schwartz, 1965))**.** *Let $D_i \overset{i.i.d.}{\sim} f_{\phi^*}$ and let the Bayesian model be*

$$\begin{aligned} D_i \mid \phi &\overset{i.i.d.}{\sim} f_\phi, \quad i = 1, \ldots, n, \\ \phi &\sim \pi. \end{aligned} \tag{A.29}$$

If the model is regular according to Definition 1, the posterior concentrates around ϕ^* as $n \to \infty$:

$$\pi\left(U^c \mid D_{1:n}\right) \overset{P_{\phi^*}}{\to} 0, \tag{A.30}$$

for every neighborhood U of ϕ^*.

E Derivation of the Bayes Factor for the Validation Threshold

Under the model described in Section 3.3 of the main paper, the marginal likelihoods under the null and alternative hypotheses are:

$$m\Big(\widehat{V}_S \mid H_0\Big) = \binom{n}{n\,\widehat{V}_S} \frac{1}{2^n}, \tag{A.31}$$

$$m\Big(\widehat{V}_S \mid H_1\Big) = \binom{n}{n\,\widehat{V}_S} \frac{B\Big(a + n\,\widehat{V}_S,\, b + n - n\,\widehat{V}_S\Big)\left(1 - F_{\mathrm{Beta}\left(a+n\,\widehat{V}_S,\, b+n-n\,\widehat{V}_S\right)}\left(\frac{1}{2}\right)\right)}{B(a,b)\left(1 - F_{\mathrm{Beta}(a,b)}\left(\frac{1}{2}\right)\right)}, \tag{A.32}$$

where the expression for $m(\widehat{V}_S \mid H_1)$ follows by integrating the binomial likelihood against the truncated Beta prior over $\left(\frac{1}{2}, 1\right)$. Taking the ratio and cancelling the binomial coefficient

yields:

$$BF_n = \frac{B\Big(a + n\,\widehat{V}_S,\, b + n - n\,\widehat{V}_S\Big)\,\Big(1 - F_{\mathrm{Beta}\left(a+n\,\widehat{V}_S,\, b+n-n\,\widehat{V}_S\right)}\left(\frac{1}{2}\right)\Big)}{B(a,b)\,\Big(1 - F_{\mathrm{Beta}(a,b)}\left(\frac{1}{2}\right)\Big)} \cdot 2^n, \tag{A.33}$$

where $B(a,b)$ is the Beta function and $F_{\mathrm{Beta}(a,b)}$ is the CDF of the $\mathrm{Beta}(a,b)$ distribution.

The distribution of BF_n under H_0 has closed form: since $n\,\widehat{V}_S \in \{0,\ldots,n\}$, the Bayes factor takes at most $n+1$ distinct values $x_0,\ldots,x_n$ with

$$\mathbb{P}(BF_n = x_k \mid H_0) = \binom{n}{k}\frac{1}{2^n}, \quad k = 0,\ldots,n, \tag{A.34}$$

where

$$x_k = \frac{B(a+k,\, b+n-k)\,\Big(1 - F_{\mathrm{Beta}(a+k,\, b+n-k)}\left(\frac{1}{2}\right)\Big)}{B(a,b)\,\Big(1 - F_{\mathrm{Beta}(a,b)}\left(\frac{1}{2}\right)\Big)} \cdot 2^n, \quad k = 0,\ldots,n. \tag{A.35}$$

The same formulas apply to the distribution of BF_n under any $V_S = v_S$, by replacing the binomial probabilities $\binom{n}{k}2^{-n}$ with $\binom{n}{k}v_S^k(1-v_S)^{n-k}$.

F Simulation Study: Data-Generating Processes and Implementation Details

Setting 1 (Useless surrogate). Potential outcomes are drawn independently from:

$$P_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}_4(\mu^*, \Sigma^*)\,, \quad \mu^* = \left(5,\, \tfrac{2}{3},\, 3,\, \tfrac{2}{3}\right)^\top, \quad \Sigma^* = \mathrm{diag}(3,\, 1,\, 3,\, 1). \tag{A.36}$$

The surrogate potential outcomes are independent of the primary outcome potential outcomes, so S carries no information about the treatment effect on Y.

Setting 2 (Perfect surrogate). The primary outcome potential outcomes $(Y_{1i}, Y_{0i}) \overset{\text{i.i.d.}}{\sim} \mathcal{N}_2\big((6, 2.5)^\top,\, 3I_2\big)$, and the surrogate is defined as $S_{gi} = Y_{gi} + \varepsilon_{gi}$, where $(\varepsilon_{1i}, \varepsilon_{0i}) \overset{\text{i.i.d.}}{\sim} \mathcal{N}_2(0,\, 0.1\, I_2)$ independently. The resulting joint distribution of $P_i = (Y_{1i}, S_{1i}, Y_{0i}, S_{0i})^\top$ is $\mathcal{N}_4(\mu^*, \Sigma^*)$ with

$$\mu^* = (6,\, 6,\, 2.5,\, 2.5)^\top, \qquad \Sigma^* = \begin{bmatrix} 3 & 3 & 0 & 0 \\ 3 & 3.1 & 0 & 0 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 3 & 3.1 \end{bmatrix}. \tag{A.37}$$

The noise is negligible ($\sigma_\varepsilon^2 = 0.1$), making S nearly identical to Y and therefore a near-perfect surrogate.

Setting 3 (Imperfect surrogate). The surrogate potential outcomes $(S_{1i}, S_{0i}) \overset{\text{i.i.d.}}{\sim} \mathcal{N}_2\big((5, 3)^\top,\, 3I_2\big)$, and the primary outcome is $Y_{gi} = S_{gi} + \varepsilon_{gi}$, where $\varepsilon_{1i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(1.5,\, 0.6)$ and $\varepsilon_{0i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\, 0.6)$ independently. The resulting joint distribution is $\mathcal{N}_4(\mu^*, \Sigma^*)$ with

$$\mu^* = (6.5,\, 5,\, 3,\, 3)^\top, \qquad \Sigma^* = \begin{bmatrix} 3.6 & 3 & 0 & 0 \\ 3 & 3 & 0 & 0 \\ 0 & 0 & 3.6 & 3 \\ 0 & 0 & 3 & 3 \end{bmatrix}. \tag{A.38}$$

The noise is more substantial ($\sigma_\varepsilon^2 = 0.6$), making S a valid but imperfect surrogate.

Setting 4 (Misspecified model). The surrogate potential outcomes are log-normal: $S_{1i} \sim \exp(\mathcal{N}(2.5,\, 1.5^2))$ and $S_{0i} \sim \exp(\mathcal{N}(0.5,\, 1.5^2))$. The primary outcome is a non-linear

function of the surrogate plus log-normal noise:

$$Y_{1i} = 2.0 + 1.2\sqrt{S_{1i}} + 0.3\exp(S_{1i}/500) + \exp(\varepsilon_{1i}), \tag{A.39}$$

$$Y_{0i} = 0.8\sqrt{S_{0i}} + 0.2\exp(S_{0i}/50) + \exp(\varepsilon_{0i}), \tag{A.40}$$

where $\varepsilon_{gi} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\, 0.3)$. This setting is designed to violate the Gaussian parametric assumptions of our Bayesian model.

Setting 5. A binary baseline covariate $X_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(0.5)$ is generated, and the distribution of $P_i \mid X_i$ is the following mixture:

$$P_i \mid X_i \overset{\text{ind.}}{\sim} X_i\, \mathcal{N}_4\big(\mu^1, \Sigma^1\big) + (1 - X_i)\, \mathcal{N}_4\big(\mu^0, \Sigma^0\big)\,, \quad i = 1, \ldots, n, \tag{A.41}$$

with

$$\mu^0 = (5, 5, 0, 0)^\top, \qquad \mu^1 = (5, -5, 0, -10)^\top,$$
$$\Sigma^0 = \Sigma^1 = \begin{bmatrix} A & 0 \\ 0 & A \end{bmatrix}, \qquad A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}. \tag{A.42}$$

Setting 6. A Gaussian baseline covariate $X_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(3, 1)$ is generated, and the distribution of $P_i \mid X_i$ follows a linear Gaussian model:

$$P_i \mid X_i \overset{\text{ind.}}{\sim} \mathcal{N}_4(X_i\, \beta,\, \Sigma)\,, \quad i = 1, \ldots, n, \tag{A.43}$$

with

$$\beta = (1,\, 7,\, 0,\, 6)^\top, \qquad \Sigma = \tfrac{1}{2}\, I_4. \tag{A.44}$$

Implementation details. For the Bayesian method, we ran $T = 500$ posterior sampling iterations with a burn-in of $b = 125$. Prior parameters were set as follows across both models:

$$\mu_0 = (0, 0, 0, 0)^\top, \qquad \Sigma_0 = 10\, I_4, \qquad s = 2, \qquad \tau = 1, \tag{A.45}$$

$$\mu_\beta = (0, 0, 0, 0)^\top, \qquad \Sigma_\beta = 10\, I_4. \tag{A.46}$$

The significance level was set to $\alpha = 0.05$. To compute the surrogate validation threshold η, we set the desired power to $1 - \beta = 0.8$ and used a uniform prior for V_S by setting $a = b = 1$.